

PRACTICAL COMPETENCY ROADMAP

# PORTFOLIO PROJECTS GUIDE

What to build, when to build it, how to test it, and what evidence to save

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# How to use this project roadmap

This guide sits beside the Weekly Study Guide and the Automation & Controls Syllabus. Study the milestone foundations first, pass the readiness gate, then begin the listed project. The dates are targets, not permission to skip a weak foundation.

## The employer test

A project counts only when you can explain it, operate it, test it, break it safely in simulation, diagnose the fault, correct it, verify it, and show the documents that define expected behaviour.

| # | Month          | Milestone                                 | Project                                                | Start                  |
|---|----------------|-------------------------------------------|--------------------------------------------------------|------------------------|
| 1 | September 2026 | Electrical and PLC Foundations            | 24 VDC Control and PLC Starter Pack                    | Saturday, 19 September |
| 2 | October 2026   | PLC Control Patterns                      | Automated Conveyor System                              | Saturday, 10 October   |
| 3 | November 2026  | Industrial Electrical and Instrumentation | Motor and Sensor Interface Mini-Project                | Saturday, 14 November  |
| 4 | December 2026  | Analog Signals and Process Control        | Automated Tank / Filling System                        | Saturday, 12 December  |
| 5 | January 2027   | HMI/SCADA and Industrial Networking       | SCADA Process Control System                           | Saturday, 9 January    |
| 6 | February 2027  | Integrated Automation System              | Automated Water Treatment Plant - Flagship             | Saturday, 6 February   |
| 7 | March 2027     | Employment Readiness                      | Application Readiness Pack and Technical Demonstration | Saturday, 6 March      |

## RULES BEFORE STARTING A PROJECT

- ☐ Complete the milestone's essential study and practical exercises.
- ☐ Write the objective, boundaries, assumptions, and acceptance criteria.
- ☐ Label the work Simulation / Training Project unless it is a genuine physical prototype.
- ☐ Pass the readiness gate without copying an answer.
- ☐ Create the first I/O, instrument, tag, or evidence list before programming.
- ☐ If the gate is not passed, delay dependent work and repair the missing foundation.

# What makes a project credible

The purpose is not to collect impressive project names. The purpose is to produce evidence of junior-level reasoning, implementation, testing, troubleshooting, and documentation.

| Weak evidence                   | Credible evidence                                                                    |
|---------------------------------|--------------------------------------------------------------------------------------|
| I watched a PLC tutorial.       | I wrote the logic, explained every rung, and recorded normal and fault tests.        |
| I know 4-20 mA.                 | I scaled signals, checked under/over-range, and documented the transmitter range.    |
| I made an HMI screen.           | I mapped tags, showed data quality, tested alarms, and handled communications loss.  |
| My project worked.              | The test sheet records expected result, actual result, status, evidence, and retest. |
| I can troubleshoot.             | A report shows symptoms, isolation tests, root cause, correction, and verification.  |
| Industrial water plant project. | Personal simulation with explicit assumptions and no false production claims.        |

## DEFINITION OF DONE FOR EVERY TECHNICAL PROJECT

- |                                                                                                                                  |                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> Objective and process are understandable in under one minute.                                           | <input type="checkbox"/> Architecture, I/O, instruments, tags, and data paths are documented.               |
| <input type="checkbox"/> PLC logic has comments and defined modes, states, permissives, interlocks, alarms, and reset behaviour. | <input type="checkbox"/> Normal operation, shutdown, faults, and recovery have expected and actual results. |
| <input type="checkbox"/> At least two realistic faults were investigated systematically.                                         | <input type="checkbox"/> Screenshots are readable and captions explain exactly what they prove.             |
| <input type="checkbox"/> Limitations and lessons learned are honest.                                                             | <input type="checkbox"/> You can demonstrate the project without reading copied notes.                      |

### Status language

Use Studied, Simulated, Implemented, Tested, and Documented precisely. Use Certified only when you possess a valid formal certificate. Use NOT TESTED until a test has actually been run.

# Recommended evidence structure

Use the same folder pattern for each project so recruiters can find evidence quickly.

| Folder / file                 | Minimum content                                                                                                    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------|
| README.md                     | Objective, project type, process, architecture, technologies, logic, tests, faults, results, limitations, lessons. |
| 01-process                    | Process overview, architecture, P&ID or control diagram.                                                           |
| 02-plc                        | Program backup/export, tag list, readable logic screenshots, software/version note.                                |
| 03-hmi-scada                  | Overview, detail, alarms, trends, diagnostics and captions.                                                        |
| 04-electrical-instrumentation | Schematics, loop/wiring diagrams, I/O and instrument lists, ranges and signals.                                    |
| 05-network                    | Architecture, IP plan, protocol/tag/register map and comms tests.                                                  |
| 06-testing                    | Test procedure, actual results, evidence references and retests.                                                   |
| 07-troubleshooting            | One report per fault with root cause and verification.                                                             |
| 08-change-log                 | Version, date, change, reason, affected tests and result.                                                          |
| 09-demo                       | Short video/GIF and a concise demonstration outline.                                                               |

## EVIDENCE NAMING EXAMPLE

P02-Conveyor\_PLC-Start-Stop\_v01\_2026-10-15.png

P04-Tank\_Test-T005\_Lost-Level-Signal\_2026-12-28.pdf

## EVIDENCE QUALITY CHECK

- |                                                                                      |                                                                                                    |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> The file opens and the important content is readable.       | <input type="checkbox"/> The caption states what the evidence demonstrates.                        |
| <input type="checkbox"/> The project, date/version, and evidence status are visible. | <input type="checkbox"/> No private licence keys, passwords, or employer information are included. |
| <input type="checkbox"/> The evidence supports the claim beside it.                  | <input type="checkbox"/> A recruiter can understand it without the original tutorial.              |

# Project 1: 24 VDC Control and PLC Starter Pack

## Milestone 1 - Electrical and PLC Foundations

September 2026 | Simulation / Training Project; optional supervised 24 VDC prototype

| Preparation    | Gate review          | Target start           | Target finish           |
|----------------|----------------------|------------------------|-------------------------|
| 7-18 September | Friday, 18 September | Saturday, 19 September | Wednesday, 30 September |

### Project objective

Prove that you understand a basic control circuit, PLC scan/I/O, and simple Ladder Logic before moving into larger machines.

## What the project is about

Start button -> control logic -> output lamp or simulated motor. Add Stop, seal-in logic, and a timed traffic-light sequence.

## Why it belongs in your portfolio

Shows that you did not skip electrical and PLC foundations. Recruiters can see that you understand inputs, outputs, control conditions, and testing.

### REQUIRED SYSTEM CONTENT

- ☐ A labelled 24 VDC control-circuit concept with supply, protection, push buttons, relay/contacter concept, load, and common/return.
- ☐ Motor or lamp Start/Stop seal-in logic with a dominant Stop condition.
- ☐ A small, consistent tag list and PLC I/O map.
- ☐ PLC hardware block diagram showing CPU, power supply, input module, output module, and field devices.
- ☐ Three-stage traffic-light sequence using timers and safe reset behaviour.
- ☐ Comments explaining what every rung does.

### SIGNALS AND CONTROL LOGIC

| Signals / field data                                                                                               | PLC and control logic                                                                 |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Start PB - DI, Stop PB - DI, E-stop healthy status - DI, Auto/Reset - DI, Run command - DO, Status lamp/alarm - DO | NO/NC logic, Seal-in circuit, Stop dominance, Timers, Basic sequence, Reset behaviour |

### Operator interface

A simple status panel or HMI sketch showing Ready, Running, Stopped, E-stop Not Healthy, and sequence step.

### Safety and credibility

Treat the E-stop as a training status/permissive only. Do not claim that ordinary PLC logic is a certified safety function.

# Project 1 execution and evidence

Target completion: Wednesday, 30 September | Do not mark work complete until the actual result is recorded.

## WHEN TO DO THE WORK

| Dates     | Stage             | Required outcome                                                                  |
|-----------|-------------------|-----------------------------------------------------------------------------------|
| 7-11 Sep  | Learn             | Electricity basics, Ohm's Law, PLC hardware, scan cycle, digital I/O.             |
| 12-18 Sep | Practise and gate | Calculations, I/O list, starter logic, traffic-light exercise, closed-book check. |
| 19-25 Sep | Build             | Complete logic, panel/HMI sketch, tags, and diagrams.                             |
| 26-30 Sep | Test and document | Run normal/fault tests, correct issues, save evidence, and sign off.              |

## ACCEPTANCE TESTS

- |                                                                             |                                                                      |
|-----------------------------------------------------------------------------|----------------------------------------------------------------------|
| <input type="checkbox"/> Start produces the intended output                 | <input type="checkbox"/> Stop removes the output                     |
| <input type="checkbox"/> Start cannot override Stop/E-stop unhealthy        | <input type="checkbox"/> Power-up does not cause an unintended start |
| <input type="checkbox"/> Traffic-light sequence and reset behave as written |                                                                      |

## FAULT-FINDING SCENARIOS

- |                                                           |                                                                  |
|-----------------------------------------------------------|------------------------------------------------------------------|
| <input type="checkbox"/> Start input not changing         | <input type="checkbox"/> Output commanded but load not indicated |
| <input type="checkbox"/> Timer sequence stuck on one step |                                                                  |

## REQUIRED DOCUMENTATION

- |                                         |                                                         |
|-----------------------------------------|---------------------------------------------------------|
| <input type="checkbox"/> Control sketch | <input type="checkbox"/> PLC block diagram              |
| <input type="checkbox"/> I/O list       | <input type="checkbox"/> Ladder Logic print/screenshots |
| <input type="checkbox"/> Test sheet     | <input type="checkbox"/> Troubleshooting report         |
| <input type="checkbox"/> Change log     |                                                         |

## PORTFOLIO EVIDENCE TO SAVE

- |                                             |                                                     |
|---------------------------------------------|-----------------------------------------------------|
| <input type="checkbox"/> One overview image | <input type="checkbox"/> Readable logic screenshots |
| <input type="checkbox"/> Completed I/O list | <input type="checkbox"/> Expected vs actual tests   |
| <input type="checkbox"/> One fault report   | <input type="checkbox"/> Short demonstration video  |

### Competence statement after completion

I built and tested basic PLC digital control logic and can explain the input-to-logic-to-output path.

## PROJECT SIGN-OFF

- |                                                                                                                           |                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> I can explain every important tag, rung, mode, permissive, interlock, alarm, and fault response. | <input type="checkbox"/> Normal, stop, abnormal, and recovery tests have honest actual results. |
| <input type="checkbox"/> Documents agree with the implemented simulation.                                                 | <input type="checkbox"/> Evidence is readable, named clearly, and labelled accurately.          |
| <input type="checkbox"/> PROJECT 1 COMPLETE - date: _____                                                                 |                                                                                                 |

# Project 2: Automated Conveyor System

## Milestone 2 - PLC Control Patterns

October 2026 | Personal Simulation / Training Project

| Preparation | Gate review       | Target start         | Target finish        |
|-------------|-------------------|----------------------|----------------------|
| 1-9 October | Friday, 9 October | Saturday, 10 October | Saturday, 31 October |

### Project objective

Demonstrate reliable PLC machine-control logic with modes, interlocks, permissives, sequencing, object detection, counting, and fault recovery.

## What the project is about

Objects enter a conveyor, are detected and counted, and move through a controlled sequence. The motor operates only when required permissives are healthy.

## Why it belongs in your portfolio

This is a strong junior PLC project because it demonstrates the logic patterns used around machines without pretending to be a production installation.

### REQUIRED SYSTEM CONTENT

- ☐ Start/Stop and E-stop healthy status with a clear Ready condition.
- ☐ Object sensor, timer, counter, target count, and counter reset.
- ☐ Interlocks and permissives separated from alarms and trips.
- ☐ Auto/Manual selector with defined command ownership.
- ☐ Motor command, run feedback simulation, overload input, and start-failure timeout.
- ☐ Fault latch, operator reset rules, and restart inhibition.

### SIGNALS AND CONTROL LOGIC

| Signals / field data                                                                                                                         | PLC and control logic                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Start, Stop, Reset - DI, E-stop healthy - DI, Object sensor - DI, Overload healthy - DI, Auto/Manual - DI, Motor command, alarm, status - DO | Timers/counters, One-shot counting, Auto/Manual, Permissives, Interlocks/trips, Command-feedback alarm |

### Operator interface

Conveyor HMI showing mode, Ready, command, running feedback, count, sensor status, overload, active fault, and reset control.

### Safety and credibility

Document the emergency-stop concept, overload response, guarded restart expectation, and why safety functions need appropriate hardware and design.

# Project 2 execution and evidence

Target completion: Saturday, 31 October | Do not mark work complete until the actual result is recorded.

## WHEN TO DO THE WORK

| Dates     | Stage                 | Required outcome                                                             |
|-----------|-----------------------|------------------------------------------------------------------------------|
| 1-9 Oct   | Learn and gate        | Timers, counters, edges, modes, permissives, interlocks, trips, fault reset. |
| 10-16 Oct | Design and core logic | Process sketch, I/O, control narrative, Start/Stop, modes, motor control.    |
| 17-23 Oct | Sequence and HMI      | Object logic, counting, alarms, status, HMI, comments.                       |
| 24-31 Oct | Faults and evidence   | Inject faults, execute tests, correct logic, publish documentation.          |

## ACCEPTANCE TESTS

- |                                                                        |                                                                               |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <input type="checkbox"/> Valid Start runs conveyor                     | <input type="checkbox"/> Stop and unhealthy E-stop stop/inhibit operation     |
| <input type="checkbox"/> Each object increments once                   | <input type="checkbox"/> Overload trips and latches alarm                     |
| <input type="checkbox"/> Fault reset works only after the cause clears | <input type="checkbox"/> Power restoration does not auto-restart unexpectedly |

## FAULT-FINDING SCENARIOS

- |                                                               |                                                          |
|---------------------------------------------------------------|----------------------------------------------------------|
| <input type="checkbox"/> Pump/motor will not start equivalent | <input type="checkbox"/> Object sensor stuck on          |
| <input type="checkbox"/> Motor overload                       | <input type="checkbox"/> Command present but no feedback |

## REQUIRED DOCUMENTATION

- |                                                  |                                                 |
|--------------------------------------------------|-------------------------------------------------|
| <input type="checkbox"/> Process diagram         | <input type="checkbox"/> I/O list               |
| <input type="checkbox"/> Control narrative       | <input type="checkbox"/> Cause & Effect         |
| <input type="checkbox"/> Alarm list              | <input type="checkbox"/> Test procedure/results |
| <input type="checkbox"/> Troubleshooting reports |                                                 |

## PORTFOLIO EVIDENCE TO SAVE

- |                                             |                                            |
|---------------------------------------------|--------------------------------------------|
| <input type="checkbox"/> PLC screenshots    | <input type="checkbox"/> I/O table         |
| <input type="checkbox"/> Conveyor diagram   | <input type="checkbox"/> HMI screenshot    |
| <input type="checkbox"/> Six recorded tests | <input type="checkbox"/> Two fault reports |
| <input type="checkbox"/> Demo video/GIF     |                                            |

### Competence statement after completion

I implemented and tested a simulated conveyor sequence with modes, permissives, interlocks, counting, trips, and controlled recovery.

## PROJECT SIGN-OFF

- |                                                                                                                           |                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> I can explain every important tag, rung, mode, permissive, interlock, alarm, and fault response. | <input type="checkbox"/> Normal, stop, abnormal, and recovery tests have honest actual results. |
| <input type="checkbox"/> Documents agree with the implemented simulation.                                                 | <input type="checkbox"/> Evidence is readable, named clearly, and labelled accurately.          |
| <input type="checkbox"/> PROJECT 2 COMPLETE - date: _____                                                                 |                                                                                                 |



# Project 3: Motor and Sensor Interface

## Mini-Project

### Milestone 3 - Industrial Electrical and Instrumentation

November 2026 | Personal Simulation / Training Project

| Preparation   | Gate review         | Target start          | Target finish       |
|---------------|---------------------|-----------------------|---------------------|
| 1-13 November | Friday, 13 November | Saturday, 14 November | Monday, 30 November |

#### Project objective

Show how motor-control equipment and field sensors interface with a PLC through commands, feedback, status, and fault signals.

### What the project is about

The PLC receives mode, permissive, sensor, run-feedback, and fault signals; it commands a contactor/VFD concept and verifies the expected response.

### Why it belongs in your portfolio

Demonstrates that PLC logic is connected to real field concepts: contactors, overloads, VFD status, sensor wiring, feedback, and electrical documentation.

#### REQUIRED SYSTEM CONTENT

- ☐ Basic motor power/control schematic showing protection, contactor or VFD concept, overload, and motor.
- ☐ Local/Remote concept and defined remote-control permissive.
- ☐ Run, Ready, At Speed, Fault, and Reset signal map as applicable.
- ☐ PLC command-to-feedback logic with a configurable start-confirmation delay.
- ☐ Two sensors with documented PNP/NPN and wiring assumptions.
- ☐ Alarm/status indication and a safe response to loss of feedback.

#### SIGNALS AND CONTROL LOGIC

| Signals / field data                                                                                                                          | PLC and control logic                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Remote selected - DI, Drive/contactor ready - DI, Run feedback - DI, Fault/overload - DI, Two sensor inputs - DI, Run and reset commands - DO | Ready permissive, Start command, Feedback timer, Failed-to-start alarm, Fault latch/reset, Local/Remote handling |

#### Operator interface

Motor faceplate or status panel showing Ready, Commanded, Running, At Speed, Faulted, Local/Remote, and sensor states.

#### Safety and credibility

Do not perform unsupervised mains or three-phase wiring. A physical version should remain at safe extra-low voltage and use appropriate supervision.

# Project 3 execution and evidence

Target completion: Monday, 30 November | Do not mark work complete until the actual result is recorded.

## WHEN TO DO THE WORK

| Dates     | Stage                    | Required outcome                                                          |
|-----------|--------------------------|---------------------------------------------------------------------------|
| 1-6 Nov   | Electrical concepts      | Relays, contactors, overloads, motors, three-phase awareness, schematics. |
| 7-13 Nov  | Instrumentation and gate | Sensors, PNP/NPN, wiring, VFD interface, feedback and fault concepts.     |
| 14-20 Nov | Design/build             | Schematic, I/O, logic, motor faceplate, signal simulation.                |
| 21-30 Nov | Test/document            | Fault injection, normal tests, wiring assumptions, reports, evidence.     |

## ACCEPTANCE TESTS

- |                                                                |                                                                  |
|----------------------------------------------------------------|------------------------------------------------------------------|
| <input type="checkbox"/> Motor starts only when Ready          | <input type="checkbox"/> Local mode blocks remote command        |
| <input type="checkbox"/> Missing feedback creates an alarm     | <input type="checkbox"/> Fault stops/inhibits the command        |
| <input type="checkbox"/> Reset is rejected while fault remains | <input type="checkbox"/> Sensor states map correctly to PLC tags |

## FAULT-FINDING SCENARIOS

- |                                                                     |                                               |
|---------------------------------------------------------------------|-----------------------------------------------|
| <input type="checkbox"/> Motor overload                             | <input type="checkbox"/> Run feedback missing |
| <input type="checkbox"/> Sensor open circuit/wrong polarity concept | <input type="checkbox"/> VFD not ready        |

## REQUIRED DOCUMENTATION

- |                                                       |                                                |
|-------------------------------------------------------|------------------------------------------------|
| <input type="checkbox"/> Electrical/control schematic | <input type="checkbox"/> Wiring/loop sketch    |
| <input type="checkbox"/> I/O list                     | <input type="checkbox"/> Motor/VFD signal list |
| <input type="checkbox"/> Control narrative            | <input type="checkbox"/> Test sheet            |
| <input type="checkbox"/> Fault report                 |                                                |

## PORTFOLIO EVIDENCE TO SAVE

- |                                                   |                                                                |
|---------------------------------------------------|----------------------------------------------------------------|
| <input type="checkbox"/> Schematic screenshot/PDF | <input type="checkbox"/> Signal map                            |
| <input type="checkbox"/> PLC interface logic      | <input type="checkbox"/> Faceplate/status view                 |
| <input type="checkbox"/> Test results             | <input type="checkbox"/> One motor and one sensor fault report |

### Competence statement after completion

I can trace a motor or sensor signal from the field concept through PLC I/O and logic to operator status and fault response.

## PROJECT SIGN-OFF

- |                                                                                                                           |                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> I can explain every important tag, rung, mode, permissive, interlock, alarm, and fault response. | <input type="checkbox"/> Normal, stop, abnormal, and recovery tests have honest actual results. |
| <input type="checkbox"/> Documents agree with the implemented simulation.                                                 | <input type="checkbox"/> Evidence is readable, named clearly, and labelled accurately.          |
| <input type="checkbox"/> PROJECT 3 COMPLETE - date: _____                                                                 |                                                                                                 |

# Project 4: Automated Tank / Filling System

## Milestone 4 - Analog Signals and Process Control

December 2026 | Personal Simulation / Training Project

| Preparation   | Gate review         | Target start          | Target finish                               |
|---------------|---------------------|-----------------------|---------------------------------------------|
| 1-11 December | Friday, 11 December | Saturday, 12 December | Thursday, 31 December<br>(holiday-flexible) |

### Project objective

Demonstrate 4-20 mA scaling, multiple process measurements, pump/valve control, alarms, interlocks, Auto/Manual modes, and process documentation.

## What the project is about

Inlet valve -> tank -> pump -> outlet valve. Level is the primary control variable, with simulated pressure, temperature, and flow measurements.

## Why it belongs in your portfolio

This connects PLC logic to instrumentation and process behaviour, which is essential for controls, instrumentation, water, mining, and process-industry roles.

### REQUIRED SYSTEM CONTENT

- |                                                                                                                           |                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> Level, pressure, temperature, and flow transmitter simulations with documented ranges and units. | <input type="checkbox"/> 4-20 mA/raw-value scaling and under-range/over-range fault detection.      |
| <input type="checkbox"/> Inlet valve, pump, and outlet valve controls with position/run feedback concepts.                | <input type="checkbox"/> Auto filling sequence and controlled Manual operation.                     |
| <input type="checkbox"/> High alarm, high-high trip/shutdown, low-level pump protection, and reset logic.                 | <input type="checkbox"/> Basic feedback-control concept with PV, SP, manipulated output, and trend. |

### SIGNALS AND CONTROL LOGIC

| Signals / field data                                                                                                                             | PLC and control logic                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| LT, PT, TT, FT - AI, Valve feedback - DI, Pump ready/run/fault - DI, Valve commands - DO, Pump command - DO, Optional speed/valve reference - AO | Analog scaling, Plausibility/fault checks, Auto sequence, Modes, Alarms/trips, Basic feedback/PID concept |

### Operator interface

Tank overview with PVs and units, fill level, pump/valve status, mode, SP, alarms, trends, signal quality, and communication status.

### Safety and credibility

High-high response and low-level pump protection are simulated control concepts, not a certified safety instrumented function.

# Project 4 execution and evidence

Target completion: Thursday, 31 December (holiday-flexible) | Do not mark work complete until the actual result is recorded.

## WHEN TO DO THE WORK

| Dates     | Stage             | Required outcome                                                                           |
|-----------|-------------------|--------------------------------------------------------------------------------------------|
| 1-11 Dec  | Learn and gate    | 4-20 mA, scaling, calibration, valves, PV/SP, feedback, PID fundamentals.                  |
| 12-18 Dec | Design/build      | P&ID, ranges, I/O, instruments, scaling, equipment logic.                                  |
| 19-24 Dec | Integrate         | Auto/Manual, alarms, trips, HMI, trends, control narrative.                                |
| 25-31 Dec | Light test/finish | Holiday-flexible testing, documentation and evidence; extend into early January if needed. |

## ACCEPTANCE TESTS

- |                                                                 |                                                                                  |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------|
| <input type="checkbox"/> 4 mA maps to LRV and 20 mA maps to URV | <input type="checkbox"/> Auto fill stops at the intended condition               |
| <input type="checkbox"/> Low level protects the pump            | <input type="checkbox"/> High-high initiates the documented shutdown             |
| <input type="checkbox"/> Bad analog signal alarms clearly       | <input type="checkbox"/> Manual commands remain subject to protective interlocks |

## FAULT-FINDING SCENARIOS

- |                                                    |                                                 |
|----------------------------------------------------|-------------------------------------------------|
| <input type="checkbox"/> Level transmitter failure | <input type="checkbox"/> Incorrect flow reading |
| <input type="checkbox"/> Pump will not start       | <input type="checkbox"/> Valve will not open    |
| <input type="checkbox"/> High-high tank level      |                                                 |

## REQUIRED DOCUMENTATION

- |                                            |                                                           |
|--------------------------------------------|-----------------------------------------------------------|
| <input type="checkbox"/> P&ID              | <input type="checkbox"/> I/O list                         |
| <input type="checkbox"/> Instrument list   | <input type="checkbox"/> Loop diagram                     |
| <input type="checkbox"/> Control narrative | <input type="checkbox"/> Alarm list                       |
| <input type="checkbox"/> Cause & Effect    | <input type="checkbox"/> Test and troubleshooting reports |

## PORTFOLIO EVIDENCE TO SAVE

- |                                              |                                                    |
|----------------------------------------------|----------------------------------------------------|
| <input type="checkbox"/> P&ID PDF            | <input type="checkbox"/> Scaling calculations      |
| <input type="checkbox"/> PLC screenshots     | <input type="checkbox"/> HMI and trend screenshots |
| <input type="checkbox"/> Alarm configuration | <input type="checkbox"/> Six tests                 |
| <input type="checkbox"/> Two fault reports   |                                                    |

### Competence statement after completion

I implemented and tested a simulated tank process using 4-20 mA scaling, equipment control, alarms, interlocks, and process documentation.

## PROJECT SIGN-OFF

- |                                                                                                                           |                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> I can explain every important tag, rung, mode, permissive, interlock, alarm, and fault response. | <input type="checkbox"/> Normal, stop, abnormal, and recovery tests have honest actual results. |
| <input type="checkbox"/> Documents agree with the implemented simulation.                                                 | <input type="checkbox"/> Evidence is readable, named clearly, and labelled accurately.          |
| <input type="checkbox"/> PROJECT 4 COMPLETE - date: _____                                                                 |                                                                                                 |

# Project 5: SCADA Process Control System

## Milestone 5 - HMI/SCADA and Industrial Networking

January 2027 | Personal Simulation / Training Project

| Preparation               | Gate review       | Target start        | Target finish      |
|---------------------------|-------------------|---------------------|--------------------|
| 1-8 January (light start) | Friday, 8 January | Saturday, 9 January | Sunday, 31 January |

### Project objective

Show the complete data path from field device and PLC tag through an industrial network to an operator screen, alarm, trend, and historical-data concept.

## What the project is about

Field devices -> PLC -> Ethernet/industrial protocol -> HMI/SCADA. Reuse the tank process so the focus stays on communications and operator information.

## Why it belongs in your portfolio

Proves that you understand SCADA as an operational system rather than a collection of attractive screens.

### REQUIRED SYSTEM CONTENT

- ☐ Architecture diagram showing field, PLC, switch/network, HMI/SCADA, and historian concept.
- ☐ Tag/register map tracing each displayed value to its PLC source.
- ☐ Alarm priorities, acknowledgement, timestamps, and operator-response notes.
- ☐ IP address plan and documented device ownership.
- ☐ Overview, detail/faceplate, alarm, trend, and communications-diagnostic screens.
- ☐ Watchdog/data-quality handling for PLC or SCADA communication loss.

### SIGNALS AND CONTROL LOGIC

| Signals / field data                                                                                      | PLC and control logic                                                                              |
|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Process PVs, Commands, Equipment feedback, Alarm/trip bits, Mode/status, Heartbeat/watchdog, Data quality | PLC-HMI tag mapping, Command validation, Watchdog, Comms alarm, Bad-quality display, Safe recovery |

### Operator interface

A restrained industrial screen set: overview first, details on demand, clear abnormal states, units, navigation, alarms, trends, and diagnostics.

### Safety and credibility

HMI controls must not bypass PLC permissives or interlocks. Displayed E-stop and safety status is indication only unless designed with certified systems.

# Project 5 execution and evidence

Target completion: Sunday, 31 January | Do not mark work complete until the actual result is recorded.

## WHEN TO DO THE WORK

| Dates     | Stage                 | Required outcome                                                       |
|-----------|-----------------------|------------------------------------------------------------------------|
| 1-8 Jan   | Learn and gate        | HMI/SCADA architecture, alarms, trends, Ethernet, IP, Modbus, OPC UA.  |
| 9-15 Jan  | Architecture/data     | Network diagram, IP plan, PLC tags, register/tag map, watchdog.        |
| 16-22 Jan | Build screens         | Overview, faceplates, alarms, trends, navigation, data quality.        |
| 23-31 Jan | Comms faults/evidence | Lose/recover communications, verify commands, capture tests and video. |

## ACCEPTANCE TESTS

- |                                                                           |                                                                       |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <input type="checkbox"/> Displayed values match PLC source tags           | <input type="checkbox"/> Commands reach the correct PLC tags          |
| <input type="checkbox"/> PLC loss creates bad quality/comms alarm         | <input type="checkbox"/> Recovery does not create an unintended start |
| <input type="checkbox"/> Alarms include correct state and acknowledgement | <input type="checkbox"/> Trends reveal a simulated process change     |

## FAULT-FINDING SCENARIOS

- |                                                      |                                                      |
|------------------------------------------------------|------------------------------------------------------|
| <input type="checkbox"/> PLC communication failure   | <input type="checkbox"/> SCADA communication failure |
| <input type="checkbox"/> Incorrect IP/subnet         | <input type="checkbox"/> Wrong register/tag mapping  |
| <input type="checkbox"/> Stale data/watchdog failure |                                                      |

## REQUIRED DOCUMENTATION

- |                                                       |                                           |
|-------------------------------------------------------|-------------------------------------------|
| <input type="checkbox"/> Architecture diagram         | <input type="checkbox"/> Network diagram  |
| <input type="checkbox"/> IP plan                      | <input type="checkbox"/> Tag/register map |
| <input type="checkbox"/> Alarm list                   | <input type="checkbox"/> Screen list      |
| <input type="checkbox"/> Communication test procedure | <input type="checkbox"/> Fault report     |

## PORTFOLIO EVIDENCE TO SAVE

- |                                                      |                                            |
|------------------------------------------------------|--------------------------------------------|
| <input type="checkbox"/> Architecture image          | <input type="checkbox"/> SCADA screenshots |
| <input type="checkbox"/> Alarm and trend screenshots | <input type="checkbox"/> Tag map           |
| <input type="checkbox"/> Comms-loss test results     | <input type="checkbox"/> Fault report      |
| <input type="checkbox"/> Demo video                  |                                            |

### Competence statement after completion

I built and tested a simulated PLC-to-SCADA data path with clear screens, alarms, trends, tag mapping, and communication-failure handling.

## PROJECT SIGN-OFF

- |                                                                                                                           |                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> I can explain every important tag, rung, mode, permissive, interlock, alarm, and fault response. | <input type="checkbox"/> Normal, stop, abnormal, and recovery tests have honest actual results. |
| <input type="checkbox"/> Documents agree with the implemented simulation.                                                 | <input type="checkbox"/> Evidence is readable, named clearly, and labelled accurately.          |
| <input type="checkbox"/> PROJECT 5 COMPLETE - date: _____                                                                 |                                                                                                 |

# Project 6: Automated Water Treatment Plant - Flagship

## Milestone 6 - Integrated Automation System

February 2027 | Personal Training / Simulation Project - not a real plant installation

| Preparation  | Gate review        | Target start         | Target finish       |
|--------------|--------------------|----------------------|---------------------|
| 1-5 February | Friday, 5 February | Saturday, 6 February | Sunday, 28 February |

### Project objective

Integrate the full junior skill set into one controlled project that can be demonstrated in approximately ten minutes.

## What the project is about

Raw Water -> Pumping -> Treatment Tank -> Filtration -> Treatment -> Storage -> Output.

## Why it belongs in your portfolio

This is the flagship proof of systems thinking: process, electrical equipment, instrumentation, PLC sequences, control, SCADA, networking, faults, tests, and documents must agree.

### REQUIRED SYSTEM CONTENT

- ☐ Modular process architecture for raw-water pumping, treatment, filtration, storage, and output.
- ☐ Defined modes, sequence states, transition conditions, permissives, interlocks, trips, reset, and restart behaviour.
- ☐ SCADA overview, area/detail screens, faceplates, alarms, trends, history concept, diagnostics, and communications status.
- ☐ Pumps, motors, valves, VFD concept, level/flow/pressure/temperature measurements, and 4-20 mA scaling.
- ☐ Basic feedback/PID concept where justified; no unnecessary control loops.
- ☐ Network architecture using clearly labelled Studied, Simulated, or Implemented technologies.

### SIGNALS AND CONTROL LOGIC

| Signals / field data                                                                                                                                         | PLC and control logic                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Multiple LT/FT/PT/TT - AI, Equipment ready/run/fault - DI, Valve position - DI, Pump/valve commands - DO, Speed/valve reference - AO, Heartbeat/comms status | Equipment modules, State sequencing, Interlocks/permissives, Analog fault handling, Alarm/trip/C&E, Network fault response |

### Operator interface

Plant overview plus area screens, equipment faceplates, alarm summary, trends, system health, network status, manual controls, and mode/sequence indication.

### Safety and credibility

State every simulation assumption. Do not claim safety PLC, SIS, ESD certification, commissioning, or production suitability.

# Project 6 execution and evidence

Target completion: Sunday, 28 February | Do not mark work complete until the actual result is recorded.

## WHEN TO DO THE WORK

| Dates     | Stage            | Required outcome                                                                                |
|-----------|------------------|-------------------------------------------------------------------------------------------------|
| 1-5 Feb   | Plan and gate    | Review architecture, modules, tag standard, documents, safety assumptions, acceptance criteria. |
| 6-12 Feb  | Core process     | P&ID, I/O, instruments, equipment modules, analog scaling, first process stage.                 |
| 13-19 Feb | Integrate        | Full sequence, alarms/trips, HMI, trends, network, remaining stages.                            |
| 20-28 Feb | FAT and evidence | Run full test pack, inject faults, retest corrections, finalise documents and demonstration.    |

## ACCEPTANCE TESTS

- |                                                                                  |                                                                                  |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <input type="checkbox"/> Each process stage starts only with correct permissives | <input type="checkbox"/> Sequence transitions match the control narrative        |
| <input type="checkbox"/> High-high and equipment trips match Cause & Effect      | <input type="checkbox"/> Analog faults create defined safe behaviour             |
| <input type="checkbox"/> Communication loss is visible and controlled            | <input type="checkbox"/> Restart after trip follows the written reset philosophy |
| <input type="checkbox"/> All documents and tags agree                            |                                                                                  |

## FAULT-FINDING SCENARIOS

- |                                                               |                                                   |
|---------------------------------------------------------------|---------------------------------------------------|
| <input type="checkbox"/> Pump failure                         | <input type="checkbox"/> Valve failure            |
| <input type="checkbox"/> Transmitter under-range              | <input type="checkbox"/> High-high storage level  |
| <input type="checkbox"/> Filter differential-pressure problem | <input type="checkbox"/> Network or SCADA failure |
| <input type="checkbox"/> Emergency-stop status active         |                                                   |

## REQUIRED DOCUMENTATION

- |                                                   |                                               |
|---------------------------------------------------|-----------------------------------------------|
| <input type="checkbox"/> P&ID                     | <input type="checkbox"/> Electrical schematic |
| <input type="checkbox"/> I/O and instrument lists | <input type="checkbox"/> Control narrative    |
| <input type="checkbox"/> Cause & Effect           | <input type="checkbox"/> Alarm list           |
| <input type="checkbox"/> Network diagram          | <input type="checkbox"/> FAT/test pack        |
| <input type="checkbox"/> Troubleshooting pack     | <input type="checkbox"/> Change log           |

## PORTFOLIO EVIDENCE TO SAVE

- |                                               |                                                      |
|-----------------------------------------------|------------------------------------------------------|
| <input type="checkbox"/> System architecture  | <input type="checkbox"/> PLC logic by module         |
| <input type="checkbox"/> HMI/SCADA screen set | <input type="checkbox"/> Complete document pack      |
| <input type="checkbox"/> Signed-off tests     | <input type="checkbox"/> At least four fault reports |
| <input type="checkbox"/> 10-minute demo video |                                                      |

### Competence statement after completion

I integrated and tested a personal water-treatment simulation that combines PLC control, instrumentation, SCADA, networking, documentation, and structured fault finding.

## PROJECT SIGN-OFF

- |                                                                                                                           |                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> I can explain every important tag, rung, mode, permissive, interlock, alarm, and fault response. | <input type="checkbox"/> Normal, stop, abnormal, and recovery tests have honest actual results. |
| <input type="checkbox"/> Documents agree with the implemented simulation.                                                 | <input type="checkbox"/> Evidence is readable, named clearly, and labelled accurately.          |
| <input type="checkbox"/> PROJECT 6 COMPLETE - date: _____                                                                 |                                                                                                 |



# Project 7: Application Readiness Pack and Technical Demonstration

## Milestone 7 - Employment Readiness

March 2027 | Portfolio and employment-preparation project

| Preparation | Gate review     | Target start      | Target finish       |
|-------------|-----------------|-------------------|---------------------|
| 1-5 March   | Friday, 5 March | Saturday, 6 March | Wednesday, 31 March |

### Project objective

Turn the completed technical work into concise, honest evidence that recruiters and interviewers can understand quickly.

## What the project is about

Evidence audit -> project READMEs -> portfolio updates -> CV alignment -> demonstration -> interview drills -> targeted applications.

## Why it belongs in your portfolio

Good work is not useful if an employer cannot verify it. This final project makes your evidence searchable, understandable, and defensible in an interview.

### REQUIRED SYSTEM CONTENT

- ☐ Final CV aimed at junior automation, controls, instrumentation, PLC, and SCADA roles.
- ☐ Portfolio evidence gallery with accurate captions and working links.
- ☐ Ten-minute flagship demonstration and two-minute personal introduction.
- ☐ One concise README for each main project using the same industrial case-study structure.
- ☐ Evidence index listing each artifact and whether it was studied, simulated, implemented, tested, or documented.
- ☐ Technical interview question bank and application tracker.

### SIGNALS AND CONTROL LOGIC

| Signals / field data                                                                | PLC and control logic                                                                                                                            |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Not applicable - this project organises evidence rather than controlling field I/O. | Explain PLC logic from your projects, Rebuild one rung/sequence from memory, Defend interlocks and reset design, Diagnose a live simulated fault |

### Operator interface

Demonstration should show process overview, PLC logic, alarms/trends, one normal test, one fault, the root cause, and the documentation that defines expected behaviour.

### Safety and credibility

Use honest language. Do not convert simulation work into claims of employment, commissioning, certification, or real-plant experience.

# Project 7 execution and evidence

Target completion: Wednesday, 31 March | Do not mark work complete until the actual result is recorded.

## WHEN TO DO THE WORK

| Dates     | Stage             | Required outcome                                                                                            |
|-----------|-------------------|-------------------------------------------------------------------------------------------------------------|
| 1-5 Mar   | Audit and gate    | Remove weak claims, missing links, unreadable screenshots, inconsistent tags, and unsupported PASS results. |
| 6-12 Mar  | Package           | Finish READMEs, evidence captions, portfolio, CV, and document links.                                       |
| 13-19 Mar | Rehearse          | Two mock interviews, troubleshooting aloud, rebuild logic, timed project demonstration.                     |
| 20-31 Mar | Apply and improve | Targeted applications, employer research, gap review, interview follow-up, no unrelated major topic.        |

## ACCEPTANCE TESTS

- |                                                                         |                                                                       |
|-------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <input type="checkbox"/> Every link opens                               | <input type="checkbox"/> Every screenshot is readable                 |
| <input type="checkbox"/> Every claim has supporting evidence            | <input type="checkbox"/> Ten-minute demo stays within time            |
| <input type="checkbox"/> You can explain every project without a script | <input type="checkbox"/> A mock fault can be diagnosed systematically |

## FAULT-FINDING SCENARIOS

- |                                                                 |                                                          |
|-----------------------------------------------------------------|----------------------------------------------------------|
| <input type="checkbox"/> Broken evidence link                   | <input type="checkbox"/> Claim not supported             |
| <input type="checkbox"/> Project cannot be explained            | <input type="checkbox"/> Inconsistent tag/document names |
| <input type="checkbox"/> Test marked PASS without actual result |                                                          |

## REQUIRED DOCUMENTATION

- |                                                 |                                                        |
|-------------------------------------------------|--------------------------------------------------------|
| <input type="checkbox"/> Final CV               | <input type="checkbox"/> Four main project READMEs     |
| <input type="checkbox"/> Evidence index         | <input type="checkbox"/> Project demonstration outline |
| <input type="checkbox"/> Interview score sheets | <input type="checkbox"/> Application tracker           |

## PORTFOLIO EVIDENCE TO SAVE

- |                                                 |                                           |
|-------------------------------------------------|-------------------------------------------|
| <input type="checkbox"/> Published portfolio    | <input type="checkbox"/> Organised GitHub |
| <input type="checkbox"/> Working document links | <input type="checkbox"/> Demo recording   |
| <input type="checkbox"/> Mock interview results | <input type="checkbox"/> Application log  |

### Competence statement after completion

I can clearly demonstrate what I personally studied, simulated, implemented, tested, documented, and troubleshot as a junior candidate.

## PROJECT SIGN-OFF

- |                                                                                                                           |                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> I can explain every important tag, rung, mode, permissive, interlock, alarm, and fault response. | <input type="checkbox"/> Normal, stop, abnormal, and recovery tests have honest actual results. |
| <input type="checkbox"/> Documents agree with the implemented simulation.                                                 | <input type="checkbox"/> Evidence is readable, named clearly, and labelled accurately.          |
| <input type="checkbox"/> PROJECT 7 COMPLETE - date: _____                                                                 |                                                                                                 |

# Competency coverage by project

| Project         | Electrical | PLC     | Instr.  | HMI     | Network | Docs    | Faults      |
|-----------------|------------|---------|---------|---------|---------|---------|-------------|
| 1. Starter      | Strong     | Basic   | Basic   | Sketch  | Basic   | Basic   | Basic       |
| 2. Conveyor     | Basic      | Strong  | Basic   | Working | Basic   | Strong  | Strong      |
| 3. Motor/sensor | Strong     | Working | Strong  | Working | Basic   | Strong  | Working     |
| 4. Tank         | Working    | Strong  | Strong  | Strong  | Working | Strong  | Strong      |
| 5. SCADA        | Basic      | Working | Working | Strong  | Strong  | Strong  | Strong      |
| 6. Water plant  | Working    | Strong  | Strong  | Strong  | Strong  | Strong  | Strong      |
| 7. Readiness    | Explain    | Explain | Explain | Explain | Explain | Present | Demonstrate |

## What this set can demonstrate

Completed honestly, this project set is strong evidence for junior and trainee automation, controls, PLC, instrumentation, SCADA, and control-systems applications. It supports junior competency; it does not replace supervised industrial experience, trade qualifications, or formal certification.

## FINAL PORTFOLIO RELEASE GATE

- ☐ All four main technical projects have objectives, architecture, logic, tests, faults, results, and lessons.
- ☐ The water-treatment project is clearly labelled as personal simulation/training.
- ☐ No planned technology is described as implemented.
- ☐ Every GitHub, document, image, and CV link works.
- ☐ The starter and motor/sensor projects support the larger projects without overwhelming the portfolio.
- ☐ No unperformed test is marked PASS.
- ☐ No safety awareness topic is described as formal certification.
- ☐ I can answer: what, why, how, use, faults, troubleshooting, documentation, and limitations.